Folding in R

Origami, Manifolds, and Dimension Reduction

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Contents

Preface

A rigid origami crease pattern generates a manifold whose true low-dimensional structure is known exactly, in closed form. Rigid folding bends the sheet along creases only; the facets stay flat. The folded surface is therefore isometric to the flat sheet, and geodesic distances measured along the surface are unchanged by folding. The flat crease pattern is not an approximation of the correct two-dimensional embedding. It *is* the correct embedding — the answer key.

That single property is what this book is built on.

Standard benchmarks for dimension reduction — the Swiss roll, the S-curve, the severed sphere — give one fixed geometry and a visual impression of success. A crease pattern gives an exact reconstruction error and a parameterised family of test problems. The fold angle θ runs continuously from a flat sheet to a fully compressed one, so every method's performance becomes a curve rather than a single number, and no configuration can be cherry-picked.

Chari and Pachter have argued that two-dimensional embeddings of high-dimensional biological data are systematically misleading. The usual rebuttal is anecdotal. On a manifold where the truth is known exactly, the distortion can be *measured* instead of argued about — and, more usefully, the evaluation metrics themselves can be checked against truth. That is the reason this book exists.

What this book is not

It is not a treatment of rigidity theory, computational origami design, or deep generative modelling. It is not a tutorial on `bookdown`, `make`, or project tooling. Where a section could not be tied back to the ground-truth claim above, it was cut rather than kept for completeness.

Impressum

Folding in R: Origami, Manifolds, and Dimension Reduction

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Source repository: https://github.com/r-heller/folding-in-r
Online edition: https://r-heller.github.io/folding-in-r/

This is a living document. The version you are reading was rendered on 2026-08-24.

Acknowledgments

This book stands on two bodies of work that developed independently and rarely cite each other: the mathematics of rigid origami, and the practice of non-linear dimension reduction. Bringing them together is the only original move here; both halves were built by others.

The R package ecosystem this book depends on — `bookdown`, `Rtsne`, `umap`, `igraph`, `vegan`, `torch` and their dependencies — is maintained largely by volunteers. The bibliography in `packages.bib` is generated automatically so that those authors are cited by name rather than thanked in the abstract.

How to use this book

Standing methodological rules

Three rules apply throughout and are not restated in each chapter.

1. **Every stochastic result is reported across at least 20 seeds.** t-SNE and UMAP are seed-dependent; a single run is an anecdote. Seeds are fixed in `_common.R` as `BENCH_SEEDS` and disclosed with every figure.
2. **Every θ-dependent result is reported as a curve, never as a single number.** Fold angle is a continuous difficulty parameter. Reporting one value invites cherry-picking, and hides exactly the regime changes the benchmark exists to expose.
3. **Every figure carries `fig.alt`.** No result is encoded by colour alone.

What you need

R 4.1 or newer and the packages pinned in `renv.lock`. Run `renv::restore()` once after cloning. There is nothing else to install: the pattern constructors, the sampler and the evaluation metrics are plain scripts in `R/`, sourced by `_common.R` when the book renders. The code that produces every result in this book is in the same repository as the book, at the same commit.

The Chapter 10 benchmark grid takes hours to compute. It is precomputed and committed under `data/processed/`; `scripts/run-benchmark-grid.R` is kept as the provenance record rather than run during the build.

How the chapters are shaped

Every body chapter uses the same nine sections, in the same order, with the same anchors. The anchors are namespaced by a short chapter mnemonic — `geom-`, `intro-`, `eval-` — because pandoc slugs headings by their text, and twelve chapters each carrying a heading called “Results” would otherwise collide and silently renumber each other’s cross-references.

```
## What this chapter answers  {#<mn>-question}
## Setup                      {#<mn>-setup}
## Background                 {#<mn>-background}
## <free-wording core>        {#<mn>-core}
## Results                    {#<mn>-results}
## Diagnostics                {#<mn>-diagnostics}
## Where it fails             {#<mn>-limits}
## Reproduce this             {#<mn>-reproduce}
## Further reading            {#<mn>-reading}
```

Only the core section is freely worded — it is the one that carries the chapter's actual argument, and calling it "Core" in every chapter would be worse than useless. Everything else is fixed, and `scripts/lint-chapters.R` fails the build on a chapter that is missing a slot, has them out of order, or anchors one to the wrong namespace.

Read it as a contract with you rather than a template for us. *Where it fails* is in every chapter because every method in this book fails somewhere, and a chapter that could not fill that section would not be finished. *Reproduce this* names the artefact and the script that produced it, so any number in the chapter can be traced to the run that generated it.

Callouts

Five callout classes are defined in `style/style.css`, each with a LaTeX fallback so they survive into the PDF.

- `boxinfo` — context a reader can skip without losing the thread.
- `boximportant` — a result that is easy to misread, or a default that will bite.
- `boxquestion` — something to try, with the answer available in the sources.
- `boxreport` — an aside on provenance, attribution or the state of the literature.
- `boxempty` — an unadorned block, no icon and no colour. It carries the suggested citation in Citing this book.

Write them as knitr block chunks:

```
\begin{boximportant}
umap::umap() defaults to preserve.seed = TRUE.
\end{boximportant}
```

Never as a raw `<div class="boxinfo">` and never as a pandoc fenced div (`::: boxinfo`). Those two forms are not what the stylesheet targets: they render as unstyled markup in HTML and disappear entirely in PDF, where the boxes are LaTeX environments declared in `style/preamble.tex`. That is how a sibling volume shipped eighty unstyled boxes.

How to read it

Part I builds the object. Part II tests four families of method against it. Part III is the benchmark proper and contains the book's main contribution — Chapter 9, where the evaluation metrics themselves are checked against exact truth. Part IV asks whether any of it changes a real decision.

Readers who only want the benchmark can start at Chapter 8 and refer back to Chapter 3 for the generative model.

Part I

Folding

Chapter 1

Introduction

Why should a crease pattern tell you anything about a gene expression matrix?

The pattern in panel A is the same however panel B is turned. What changes is how much of it any one view exposes — and a dimension-reduction method sees something closer to B than to A. The gap between the two is what the rest of this book measures.

- The ground-truth thesis: a rigid crease pattern *is* the answer key, not a proxy for it.
- What the book claims, and what it explicitly does not claim.
- The Chari & Pachter challenge to 2D embeddings, stated up front as the target.
- Why exact truth lets us evaluate the evaluation metrics, not just the methods.
- Roadmap: fold the sheet (Part I), test the methods (II), build the benchmark (III), change a decision (IV).

Chapter 2

The geometry of folding

What makes a crease pattern a valid manifold?

- Intrinsic versus ambient isometry — the distinction the whole book turns on. A rigid folding preserves the *path* metric exactly. It is not an isometry of the *ambient* metric: across a crease of dihedral angle ρ, the ratio of ambient to geodesic distance for a symmetric pair is exactly $\sin(\rho/2)$, so folding strictly contracts ambient distance for every $\theta > 0$.
- The strictness has a floor in floating point, and the chapter states it rather than leaving a reader to trip over it. Because $1-\sin(\rho/2)$ goes like $(\pi-\rho)^2$, the contraction stops being representable in double precision while the fold angle is still many orders of magnitude above machine epsilon. Compute that threshold in the chunk rather than quoting it, compare it against the smallest θ on the grid, and note what it implies for the vanishing-θ end of every curve in Part III.
- Developability and why a folded sheet has zero Gaussian curvature on facets.
- Curvature is **not** concentrated at the vertices, and the chapter argues against that reading rather than merely avoiding it — it is the natural wrong intuition. A folded flat sheet has vertex angle sums of exactly 2π, so discrete Gaussian curvature is identically zero; were it otherwise the isometry claim would collapse. What concentrates on the creases is the *extrinsic* second fundamental form, as a measure on the 1-skeleton, and the consequence that matters for manifold learning is that the **reach is exactly zero**.
- Rigid vs. non-rigid folding; why only rigid folding preserves the isometry we rely on.
- Maekawa's theorem: $|M-V| = 2$ at every interior vertex of a flat-foldable pattern.
- Kawasaki's theorem: alternating angle sums at an interior vertex are equal. Say plainly that the attribution history here is tangled: Kawasaki 1989,

Justin 1986 and Murata 1966 have no resolvable identifiers between them, and Hull's *Origametry* is the honest citation until one does.

- Boxed aside, one page, labelled background: the Huzita–Hatori axioms.
- **Proposition (the stress inversion).** Ambient-referenced stress measures reproduction of d_A; the contraction result above gives $d_A < d_U$ strictly for every $\theta > 0$; two-component PCA is the least-squares-optimal linear approximant to that same ambient configuration. Therefore stress($\mathcal{U}$) $>$ stress(PCA) — the exact answer key scores worse than a plainly wrong embedding. One line, no simulation. It arrives here rather than in Chapter 9 because it is a theorem about a definition, and spending grid cells to demonstrate a definition invites exactly the objection it should be immune to.
- Piecewise-flat surfaces sit *outside* the smoothness assumptions of manifold-learning theory. Stated as realism, not as a caveat. The transfer argument is by **disqualification**, and it asserts nothing about biology: a method that fails where an exact answer exists has no claim on a problem where none does.

Chapter 3

Folding as a generative model

How do you turn one pattern into a family of test problems?

- The Miura-ori parameterisation (a, b, α, θ).
- Uniform sampling on facets, and why naive vertex sampling biases the density.
- Noise models: ambient Gaussian, on-manifold, and outliers.
- What θ does to the k-nearest-neighbour graph, and where short-circuits appear.
- Why every result downstream is a curve over θ rather than a single number.

Part II

Methods under test

Chapter 4

Linear projections and where they break

When is a projection just a shadow?

- PCA via SVD with `prcomp`; why not `princomp` or an eigendecomposition of the covariance.
- At $\theta = 0$ two components are exact. Variance explained collapses as θ rises.
- One curve — and it motivates the whole book.
- Kernel PCA as the bridge into Part II.

Chapter 5

Geodesic methods

If the data lie on a folded sheet, how do you flatten it back out?

- Classical MDS with `cmdscale` as the baseline.
- Isomap: geodesics estimated by shortest paths on a kNN graph (`igraph::distances`).
- Locally linear embedding, and what it assumes about local neighbourhoods.
- Isomap is *built* for isometric embeddings, so the interesting result is where it breaks.
- Report the exact θ at which the kNN graph first bridges two facets — analytically and empirically.

Chapter 6

Neighbour embeddings

Why do t-SNE and UMAP disagree about the same data?

- `Rtsne` with `check_duplicates = TRUE`; document the duplicate-row failure mode.
- `umap(method = "naive")` — the default backend pulls Python `umap-learn`.
- Perplexity and `n_neighbors` sweeps.
- All results across `BENCH_SEEDS`, reported as distributions rather than means.
- Separate real disagreement from seed noise.

Chapter 7

Learned folds

Can a network learn the fold and the unfold at once?

- A `torch` autoencoder with a two-unit bottleneck. (`keras` would pull TensorFlow and break clean-clone reproducibility.)
- The only method here that learns an explicit inverse: encoder = fold, decoder = unfold.
- Reconstruction loss is the fold–unfold identity made computable.
- Compare the learned decoder against the true unfolding, not just against the input.

Part III

The benchmark

Chapter 8

Building crease-pattern manifolds

How do you build a manifold whose true geometry you already know?

- The helper API in `R/`, end to end.
- Pattern families beyond Miura-ori: Yoshimura, waterbomb.
- Product constructions: intrinsic dimension $2k$ in ambient $3k$, ground truth still exact.
- Random linear lift to arbitrary ambient dimension D.
- Why a benchmark stuck at intrinsic dimension 2 is not a serious benchmark.

Chapter 9

Ground truth and the evaluators

Your embedding looks good. Is it correct?

- Procrustes-aligned RMSE against the true unfolding as the headline metric.
- Trustworthiness, continuity, kNN preservation, Kruskal stress — each measured *against* truth.
- Where each standard metric is optimistic, and by how much.
- The metrics are kNN-based and mutually correlated; exact truth is what makes the comparison possible.
- The book's main experimental contribution.

Chapter 10

Benchmark results

Which method survives?

- The full method $\times$ pattern $\times$ θ $\times$ noise grid, at least 20 seeds per cell.
- Distributions, not means.
- State explicitly how much between-method variation falls inside within-method seed variance.
- Where Isomap wins, where it breaks, and the crossover.
- Grid is precomputed by `scripts/run-benchmark-grid.R` and committed; the script is the provenance record.

Chapter 11

Crease patterns vs. the Swiss roll

Does the new benchmark actually add anything?

- Head-to-head against Swiss roll and S-curve, same methods and same metrics.
- Method rankings that flip between benchmarks — and why.
- What the crease-pattern family surfaces that a single fixed geometry hides.
- An honest answer to "why not just use the Swiss roll?"

Part IV

Application

Chapter 12

From benchmark to decision

What happens when the data are real?

- Use the Chapter 10 grid to select a method and its hyperparameters.
- Apply that choice to a single-cell dataset.
- Compare against the field-default choice.
- Report a null result honestly if the benchmark-guided choice does not win.
- Closes with visualisation and honest-reporting standards: alt text, no colour-only encoding, seed disclosure, aspect-ratio discipline.

Appendix A

Notation

Symbol	Meaning
θ	Folding parameter; the difficulty axis. $\theta = 0$ is the flat sheet
ρ	Dihedral angle at a crease. $\rho = \pi$ when $\theta = 0$; derived from θ, not a second name for it
a, b, α	Miura-ori cell parameters
M, V	Mountain and valley creases
k	Neighbourhood size for kNN-graph methods
D	Ambient dimension after linear lift
$\mathcal{M}$	The folded manifold in $\mathbb{R}^3$
$\mathcal{U}$	The exact unfolding — ground truth in $\mathbb{R}^2$
d_A	Ambient distance, measured in $\mathbb{R}^3$ through the fold
d_U	Intrinsic (path) distance, measured along the surface — equal to Euclidean distance in $\mathcal{U}$

Appendix B

Datasets and codebook

Generated data

All crease-pattern manifolds are generated, not collected. Every dataset in Parts I–III is reproducible from a pattern specification, a fold angle, a sample size, a noise specification, and a seed.

Precomputed results

`data/processed/benchmark-grid.rds` — the Chapter 10 grid. Provenance: `scripts/run-benchmark-grid.R`.

External data

One dataset, used only in Chapter 12.

Fresh 68k PBMCs (Donor A) — peripheral blood mononuclear cells from a single healthy donor, from Zheng et al. (Zheng et al., 2017). Obtained from the 10x Genomics dataset repository, Cell Ranger 1.1.0 processing, Chromium Universal 3' chemistry.

`scripts/prepare-single-cell.R` downloads the filtered matrix, applies the preprocessing recorded in that script, and writes the artefact Chapter 12 reads. It is not run in CI and not run at render time, on the same provenance convention as the benchmark grid: run it once, commit the result, keep the script as the record of how.

Licence, and what may be redistributed

The dataset is published under **CC BY 4.0**, which permits redistribution with attribution. So the processed matrix *may* ship in this repository, and the constraint on whether it does is size rather than licence — the full filtered matrix is far too large for git, and Chapter 12 does not need it. What is committed is the reduced matrix the chapter actually reads, produced by the script above, with the reduction stated there and in the chapter.

Label provenance, and why the comparison is not circular

This matters more than it looks, and it was an open question until it was checked. Chapter 12's only external evaluation channel is the cell-type labels. If those labels had themselves been derived by clustering one of the embeddings under test, the comparison would be circular and the chapter would be measuring its own assumptions.

They were not. Zheng et al. generated reference profiles by sequencing ten bead-enriched purified subpopulations from the same donor, separately, and assigned each cell in the 68k population the identity of the purified population with which it had the highest Spearman correlation. The labels come from a separate set of experiments, and no clustering or embedding of the matrix under test enters the assignment.

Two caveats belong in the chapter rather than in a footnote, because both bound what a per-label result can mean.

- The assignment is made in gene-expression space — the same space the embeddings are computed from. It is independent of the *embedding*, which is the circularity that would matter here, but it is not an independent measurement of cell identity, and the chapter should not describe it as ground truth. The book reserves that phrase for the crease patterns, where it is earned.
- The authors flag their own assignments as unreliable in specific places: the gap between the highest and second-highest correlation is small for some cells, cytotoxic T against NK being the example they give, and the purified populations are nested — CD4+ T-helper contains all CD4+ cells — so a hand-written rule reassigns cells to the next-highest correlation within a nested family. Any per-label accuracy in Chapter 12 inherits that, and the pre-registered selection rule of Chapter 10 is what keeps it from being read as a ranking.

Glossary

Developable surface A surface that can be flattened into a plane without stretching or tearing.

Dihedral angle (ρ) The angle between the two facets meeting at a crease, measured in the ambient space. A crease in an unfolded sheet has $\rho = \pi$, not 0, and ρ decreases as the sheet folds. It is *derived* from θ and is never a synonym for it.

Folding parameter (θ) The parameter of the folding map, and the difficulty axis of the whole book. $\theta = 0$ is the flat sheet, and the pattern folds monotonically as θ grows. It is a parameter, not an angle read off a figure; the map $\rho(\theta)$ from it to the dihedral angle is stated once per pattern in Chapter 2.

Isometry, intrinsic and ambient An *intrinsic* isometry preserves distances measured **along** the surface. An *ambient* isometry preserves straight-line distance in the space the surface sits in. Rigid folding is the first and emphatically not the second: it contracts ambient distance strictly for every $\theta > 0$. Which of the two a method consumes is the distinction the book turns on, and the reason Chapter 9 reports two headline numbers rather than one.

Mountain / valley assignment The direction of each crease. Written M and V; encoded by linetype as well as colour in every figure.

Reach The largest r such that every point within r of the surface has a unique nearest point on it. Most manifold-learning guarantees are stated in terms of a positive reach; a creased surface has reach exactly zero, which is what puts these objects outside the theory rather than merely at its edge.

Rigid origami Folding in which facets stay flat and all deformation happens at creases.

Short-circuit Two points close in ambient space but far apart geodesically. The failure mode that breaks kNN-graph methods.

Colophon

Written in R Markdown and built with **bookdown** (Xie, 2026). Typeset in Inter and JetBrains Mono. Figures use the viridis plasma palette (option `"C"`), the series convention for this volume.

Interface icons are Font Awesome Free 6.5.1, vendored in `style/webfonts/` rather than loaded from a content delivery network. The icons are licensed CC BY 4.0, the fonts under the SIL OFL 1.1, and the accompanying code under the MIT License; see https://fontawesome.com/license/free.

```
#> - Session info --------------------------------------------------------------
#>  setting  value
#>  version  R version 4.6.1 (2026-06-24)
#>  os       Ubuntu 24.04.4 LTS
#>  system   x86_64, linux-gnu
#>  ui       X11
#>  language (EN)
#>  collate  C.UTF-8
#>  ctype    C.UTF-8
#>  tz       UTC
#>  date     2026-08-24
#>  pandoc   3.8.3 @ /opt/hostedtoolcache/pandoc/3.8.3/x64/ (via rmarkdown)
#>  quarto   NA
#>
#> ------------------------------------------------------------------------------
```

About the author

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Citing this book

This book is published under CC BY 4.0, which asks for attribution and nothing else. Quote it, teach from it, adapt it, use its figures — the licence permits all of that, and a citation is the whole of what is expected in return.

Cite the book itself for the argument and the benchmark design. Cite the repository when what you are relying on is the code: the pattern constructors, the folding maps, the metrics, or a committed artefact under `data/processed/`. They carry the same version number, and the repository is the only one of the two that pins a commit.

> Heller, R. (2026). *Folding in R: Origami, Manifolds, and Dimension Reduction.* Version 0.1.0. https://r-heller.github.io/folding-in-r/

The BibTeX entry is in `citation.bib`, and `CITATION.cff` carries the same metadata in a form GitHub and Zenodo read directly:

```
@book{heller2026folding,
  title     = {Folding in R: Origami, Manifolds, and Dimension Reduction},
  author    = {Heller, R.},
  year      = {2026},
  publisher = {Self-published},
  url       = {https://r-heller.github.io/folding-in-r/},
  note      = {Version 0.1.0}
}
```

Both files are generated from the same source and should not be edited by hand; if a version number here disagrees with one of them, the files are right.

References

Bibliography

Xie, Y. (2026). *bookdown: Authoring Books and Technical Documents with R Markdown*. R package version 0.47.

Zheng, G. X. Y., Terry, J. M., Belgrader, P., Ryvkin, P., Bent, Z. W., Wilson, R., Ziraldo, S. B., Wheeler, T. D., McDermott, G. P., Zhu, J., Gregory, M. T., Shuga, J., Montesclaros, L., Underwood, J. G., Masquelier, D. A., Nishimura, S. Y., Schnall-Levin, M., Wyatt, P. W., Hindson, C. M., Bharadwaj, R., Wong, A., Ness, K. D., Beppu, L. W., Deeg, H. J., McFarland, C., Loeb, K. R., Valente, W. J., Ericson, N. G., Stevens, E. A., Radich, J. P., Mikkelsen, T. S., Hindson, B. J., and Bielas, J. H. (2017). Massively parallel digital transcriptional profiling of single cells. *Nature Communications*, 8:14049.